LabAssist

Project Tasks & Issues

Database Stage 2 · PostgreSQL 17 · 29 October 2026

Project → Tasks → Assigned student
Project → Issues → Resolving teammate
Tasks / Issues → Append-only activity history

Tables

Table	Fields
project_tasks	id, seed_key, project_id, created_by, assigned_to, updated_by, title, description, due_on, status, needs_help, completed_at, revision, timestamps
project_issues	id, seed_key, project_id, created_by, updated_by, title, description, priority, status, resolution, resolved_by, resolved_at, revision, timestamps
project_work_events	id, project_id, actor_id, task_id or issue_id, event_key, before_state, after_state, created_at

Derived views

project_member_work_summary: total, done, in progress, remaining, overdue and personal progress.
project_team_work_summary: mean member progress, open tasks and open issues. No static completion percentages are stored.

Shared identities

References existing accounts, project_students and projects. Students are equal teammates. No Leader/Member role or project code is introduced.

Permissions & lifecycle

- Any current teammate can create a TODO task for any current teammate.
- Only the assigned student can update that task, confirm completion or request help.
- Completion records a timestamp and clears help; correcting completion clears the timestamp.
- Any current teammate can create/update/resolve project issues.
- A resolved issue requires a resolution note and records who resolved it and when.
- Supervisors do not mutate student work in this stage; backend read access follows project_supervisors.
- Project/task identity and history cannot be rewritten.
- Row-locking status functions reject stale expected revisions. This is not evidence of a tested multi-connection race.

Backend boundary

The authenticated backend must derive actor identity from the session, scope reads and apply expected revisions to edit endpoints. No client database credentials, RLS claim or live API is provided. Task reassignment/deletion and broader lifecycle permissions are not implemented in this stage.

Seed

Fixture	Count
Existing projects / students	23 / 54
Planned TODO tasks	54; one per demo student
Synthetic OPEN issues	3
Initial history records	57
Fake completed work / passwords	None

Install & verification

Run shared Stage 1 SQL 01-06, then 07-project-work-schema.sql and 08-demo-seed.sql. Existing databases apply missing stages only. Schema 07 runs once. Seeds may replay. Combined SQL is for a fresh installation only.

Docker

Disposable port 5436, database labassist_project_work. Initialize-Local.ps1 creates ignored local credentials. Docker Compose loads dependencies and this stage in order.

Acceptance

28 ordinary SQL checks passed locally in PGlite: membership, assignee-only updates, state transitions, resolution metadata, computed progress, append-only history, stale revision rejection and rollback. Shared regression: 37 student, 16 supervisor and 21 workshop checks passed.

GitHub Actions separately validates on real PostgreSQL 17. Independent-connection concurrency tests and backend/API permission tests remain pending and must not be reported as completed.

Next stages

BOM and expenses; report linkage; authenticated API and Mobile integration. Existing frontend screens remain local/mock.